

DOPE: DOWNSTREAM OBJECTIVE-PRESERVING ENCODING FOR SYNTHETIC TABULAR DATA

Anonymous

Affiliation withheld for review.

ABSTRACT

We introduce **Downstream Objective-Preserving Encoding (DOPE)**, a measurement of how much of a real-trained regressor’s improvement over a null is recovered by an auditor trained on synthetic rows. Each method is selected on its own objective. The shared column is retention, not that native score. On 97 informative PMLB regression lineages, fit seed 11 and synthetic size $4n$, the CatBoost median retention is 0.940 (95% lineage-bootstrap interval 0.914 to 0.976). Holm-adjusted Wilcoxon tests separate this generator from a Gaussian copula (0.656; 86/0/11), a Chow–Liu tree (0.586), and independent marginals (-0.012). The linear auditor does not separate it from the copula (49/0/43; Holm 0.373). That interval contains zero, and the result is not an equivalence test. Kumo large, Mitra, and TabICL repeat the same density block and are not pooled with CatBoost. MFS-v2 and MFS-v3, the gated geometric release scores, are null on every method. Official tests were not opened. The distance ledger is not differential privacy or HIPAA de-identification. The 10,240-byte cap is a release gate. A retention comparison does not pass it.

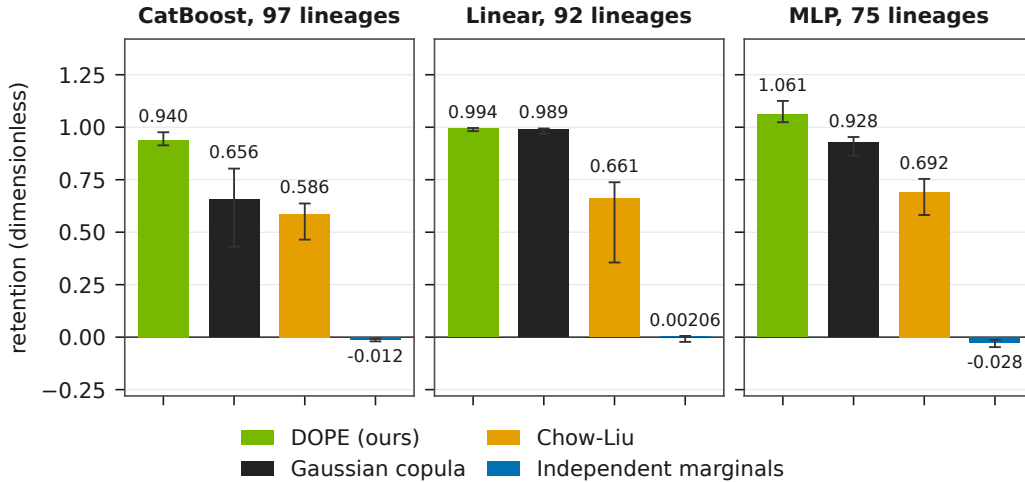


Figure 1: Median retention at synthetic size $4n$ on the density block, for CatBoost, the linear auditor, and the MLP. Each bar is a paired-lineage median and the whiskers are 95% lineage-bootstrap intervals. Under CatBoost, DOPE sits above the Gaussian copula (0.656; 86/0/11), the Chow–Liu tree (0.586), and independent marginals (-0.012). The linear auditor does not separate DOPE from the copula (49/0/43; Holm 0.373), and that result is not an equivalence. The MLP panel is the same block and is not pooled with CatBoost. Forest-Flow is a separate block and is not drawn. MFS-v2 and MFS-v3 stay null, and a retention does not pass the 10,240-byte gate.

1 INTRODUCTION

A generator paper often reports the metric that also selected the configuration. Those native scores live on different scales: predictive efficacy for CTGAN and TVAE (Xu et al., 2019), held-out log density for a Gaussian copula (Patki et al., 2016), a forest density for ARF (Watson et al., 2023), and a diffusion or CART objective for other families (Kotelnikov et al., 2023; Nowok et al., 2016). This paper separates selection from the common measurement. After each method has been selected on its own objective, the shared columns are retention and the bytes charged to the artifact.

1.1 DOWNSTREAM OBJECTIVE-PRESERVING ENCODING

We introduce **Downstream Objective-Preserving Encoding (DOPE)**, a measurement of how much of a real-trained regressor’s improvement over a null is recovered by an auditor trained on synthetic rows. The decision those columns can support is narrow. Retention is the measurement this paper can finish. It is not a declaration that one generator is best, and it is not a license to ship. Master Fitness v2 (MFS-v2) and MFS-v3, defined in Section 2.4, are the release scores. Both are null on every method here. A missing component or a failed gate removes the scalar rather than publishing a small number.

1.2 CONTRIBUTIONS

1. **Displayed residual and sampling procedure.** The displayed residual and the generative procedure that samples a row from that artifact (Section 2, Fig. 2, (1)). That procedure is not the production default and it does not use the 4,168-dimensional router.
2. **Retention with lineage-bootstrap intervals.** Retention on the validation panel, with lineage-bootstrap intervals, and the algebra of a retention value (Fig. 3). Charged bytes are the compactness check in Fig. 5, not the result.
3. **Paired tests inside unpooled blocks.** Paired tests inside unpooled blocks: the density block, the neural block, and the 6-lineage Forest-Flow confirmation, including the linear non-separation and the Forest-Flow lead (Fig. 4).
4. **A release score that stays null when a gate is missing.** The release score in Section 2.4. It is a geometric mean that stays null when any hard gate or positive-weight component is missing, so a retention cannot be copied into (3).

Scope. Official tests stay sealed. MFS-v2, MFS-v3, and PTF-v1 are null. PTF-v1 is the release KPI whose hard gates include joint fidelity, membership and attribute attacks, copy checks, and feature-importance agreement on a sealed holdout. TabSyn is measured on a retained validation subset. The 100-lineage TabDDPM publication is not measured. The full Forest-Flow publication is not measured. The pre-specified privacy-attack panel is not measured. Fit-seed variance is not measured for the displayed DOPE generator. The full ablation grid is not measured. A release at the 10,240-byte tier is a separate claim from the byte cap, and it is not made here. Section 7 records what these limits do to the estimates. An unexecuted method is not a win for DOPE.

2 METHOD

2.1 DISPLAYED GENERATOR

The displayed profile is the generator this paper reports. Its configuration id is `features12_steps2048`: at most 12 screened inputs, a hidden width of 16, a tanh activation, 2,048 AdamW steps at learning rate 0.002, a refit readout, and no embedding. It is a compact neural residual, not a GAN, a diffusion model, or a transformer. Fig. 2 is the pipeline. Training screens the inputs, maps them through the width-16 tanh layer, and runs AdamW on the regression mean-squared error, with gradient clipping at 5. The hidden weights and biases are then frozen. The intercept, the linear skip, and the hidden readout are refit on those frozen features. Coefficients whose absolute value is below 10^{-5} are dropped from the skip. For a selected input vector x_S ,

$$h_j = \tanh(b_j + w_j^\top x_S), \quad \hat{y} = a + \beta^\top x_S + v^\top h. \quad (1)$$

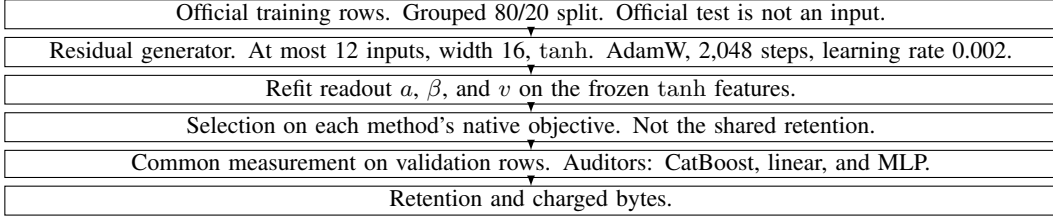


Figure 2: **Displayed pipeline.** The official test is not an input.

There is no embedding table in (1). A separate dataset-action router uses a vector of dimension $856 + 24 \times (128 + 10) = 4,168$. That vector does not generate the rows in this paper.

Equation (1) is the conditional mean, not the joint draw. Sampling for this profile uses the symbolic kernel stored in the artifact. It does not use the separate neural-joint sampler. The dependence stored with an artifact is one choice from the compile search: independence, a Chow–Liu tree, a triangular autoregression, a sparse Gaussian copula, a factor model, a truncated vine, or a two-component mixture. This paper does not treat the displayed artifacts as one dependence family.

Training fits the residual. Sampling draws a row from the stored artifact. Training, not sampling, screens inputs by absolute rank correlation with the target and keeps at most 12. The artifact stores a marginal and a transform for each feature, the selected dependence, the residual, a noise model, and a per-feature missingness rate. A draw samples a latent vector from the stored dependence, maps each coordinate through the standard normal distribution function, inverts the marginal, and inverts the feature transform. The residual then supplies $\hat{y}$. When the training target is not entirely zeros and ones, the stored noise is heteroscedastic: a linear fit of the log absolute residual on up to four screened features, with the scale clamped between the lower and upper deciles of those absolute residuals. The sampled target adds Gaussian noise of that scale and then clamps the sum to the unit interval of the training transform. It is not returned in the original units of the source column. Each feature is then replaced, independently, by a missing mark with its stored probability. The charged byte count is the length of the encoded artifact. Before that encoding, the compiler quantizes the marginals, the dependence, the residual, and the noise model.

The profile was fixed before the shared retention was inspected. Compile-time search records the proxy $0.45u + 0.25d + 0.30j$ and then keeps the candidate with the smallest normalized gate shortfall, then the smallest artifact, then the lowest candidate id. Every candidate record withholds a release certificate. The compile record is not a release certificate. No global DOPE family has been selected. The production default, used when this research profile is not selected, clamps the hidden width from 4 to 12 and trains for 96 steps. It is not this result. Other shapes exist and are not this measurement: micro-TVAEs, TinyMAT, joint TVAE, a masked transformer, and a TabSyn candidate. A direct-rank TabDDPM candidate is a research comparator with no claim on the 10,240-byte tier. The scored runs did not store a per-step loss. A later logged replay did. The supplement plots that replay. The replay files are not substituted for the scored artifacts, and a lower hidden-basis loss at 8,192 steps does not replace the displayed profile. The full width-and-step grid is not measured.

2.2 A FIXED ENCODING OF THE COLUMNS

The displayed generator and the dataset encoding both reduce a table to a fixed set of columns. The budgets differ, and both budgets belong to the current method.

Training screens columns by absolute rank correlation with the target and keeps at most 12. Those columns are the support x_S in (1). The artifact stores their positions in the training table. That screen is what keeps the residual inside the byte cap. It is a weak dataset key. A column the target does not rank is dropped, and a permutation of the remaining columns changes the stored positions.

The encoding an index can store is the sketch the router already consumes. `embed-dataset` min-max normalizes each training column onto the unit interval, leaves a missing entry missing, and does not read the official test. Each column becomes twelve content summaries: a missing rate, moments, quantiles, a unique-value share, a tail ratio, and the absolute association with the target.

Columns are sorted by those summaries, variance first, then by the rest of the summary, then by the sorted observations. A permutation of columns or of rows does not move the vector. The sketch keeps the first 64 columns in that order and pads a shorter table with zeros. Absolute correlations among the kept columns collapse to one six-number distribution, so the width does not grow with the square of the column count. A ten-number header records the task, the log row count, the log column count, and the shape of the target. The width of that sketch is 856.

The 4,168-dimensional vector appends 24 candidate states from one router bundle onto that sketch. A second bundle moves the same table. An index stores the 856-wide sketch. The longer vector is a route from one bundle.

A neighbor found with the sketch is still fit with the displayed profile. The artifact is still charged in bytes, and it still faces the release score in Section 2.4. The sketch is not a seventh fitness component. A router action that names a retention or a failure probability is a prediction from the bundle. It does not enter (3).

2.3 RETENTION

Let L_{null} , L_{real} , and $L_{\text{synthetic}}$ be the auditor’s loss on held-out validation rows when that auditor is trained on a null baseline, on real fit rows, and on synthetic rows. Those held-out rows come from the official training file. Retention is

$$\frac{L_{\text{null}} - L_{\text{synthetic}}}{L_{\text{null}} - L_{\text{real}}}. \quad (2)$$

Rearrangement, with r the retention, is algebra on those three losses:

$$L_{\text{synthetic}} = L_{\text{real}} + (1 - r)(L_{\text{null}} - L_{\text{real}}).$$

Invented losses 1, 0.01, and retention 0.94 give a synthetic-trained loss of 0.0694, which is 6.94 times the real-trained mean squared error. The CatBoost median in this paper is not this example. A value of 1 means the synthetic-trained auditor matched the real-trained auditor. A value above 1 means it beat that auditor. A negative value means it did worse than the null. A lineage is informative when the real improvement is at least 0.01 times the absolute null loss and is not zero. Otherwise the lineage is omitted. It is not entered as zero. The three auditors are CatBoost (Prokhorenkova et al., 2018), ordinary linear regression, and an MLP. The lineage score is the median of sample seeds 101, 211, and 307. The panel score is the median of those lineage scores. Intervals below are percentile lineage-bootstrap intervals, 10,000 resamples of lineages, with the three seeds kept inside the lineage median. They are not per-seed whiskers.

2.4 THE RELEASE SCORE

The decision this project is built to make is whether an artifact may ship. Retention is the measurement this paper can finish. It is the shared column after each method has been selected on its own objective. A high retention means an auditor trained on synthetic rows recovered the improvement that the same auditor obtained from real fit rows. That can be true of an artifact that is too large to ship, that misses the joint distribution, that copies training rows, or that moves the features a later model relies on. The release score exists so those failures are not traded against a utility median.

MFS-v2 is that score. With component $c_i \in [0, 1]$, weight $w_i > 0$, $\epsilon = 10^{-6}$, and $W = \sum_i w_i$,

$$\text{MFS-v2} = 100 \exp\left(\frac{1}{W} \sum_i w_i \ln(\epsilon + c_i)\right) \quad (3)$$

when the hard-gate map is non-empty, every hard gate passed, and every positive-weight component is finite and inside $[0, 1]$. Otherwise the scalar is null. The mean is geometric. A component near zero pulls the product down. A component that is missing, or that falls outside $[0, 1]$, removes the score rather than publishing a small number. An empty gate map, or any failed hard gate, does the same. Privacy has soft weight zero. A membership failure, an attribute failure, or a copied row is a gate. Utility cannot offset it. The weights are utility transfer 0.30, driver fidelity 0.20, distribution fidelity 0.20, structure fidelity 0.15, coverage realism 0.10, and compactness 0.05.

The six components ask six different questions. Utility transfer is a one-sided lower bound on retention, clamped into $[0, 1]$. It is the bound inside (3), and it is distinct from the median of (2) in the figures below. Driver fidelity is an equal mixture of feature-importance rank agreement and top-feature-set overlap between the real table and the synthetic table. Distribution fidelity is half the minimum marginal fidelity, plus a quarter of the sliced-Wasserstein fidelity and a quarter of the maximum mean discrepancy fidelity. Structure fidelity is the minimum dependence fidelity. Coverage realism, on a fixed embedding of holdout rows, is 0.25 precision, 0.25 recall, 0.20 density capped at 1, and 0.30 coverage. Compactness is $\text{clamp}(1 - \text{bytes}/10240, 0, 1)$. An artifact past 10,240 bytes fails the byte gate, and the scalar stays null. Crossing the cap is a failed gate, distinct from a retention of zero.

MFS-v3 is the score a later release has to pass. MFS-v2 is unchanged. Historical receipts still pin the v2 file. (3) is the same geometric mean, with one added weight. Utility transfer is 0.35. Representation closeness, driver fidelity, and distribution fidelity are each 0.15. Structure fidelity is 0.10. Coverage realism and compactness are each 0.05. Privacy stays at soft weight zero. A missing component, a failed gate, or any other weight vector makes the scalar null.

Distribution fidelity, structure fidelity, coverage realism, and compactness keep their v2 definitions. The driver term stored beside this panel is the same equal mixture of rank agreement and top-feature-set overlap, applied to absolute correlations with the target. It is not the permutation-importance auditor in the v2 definition, and this paper does not publish it as a component value. v2 utility transfer remains the one-sided lower bound on retention for the CatBoost, linear, and MLP auditors in the tables below. v3 utility transfer is that same bound on the Kumo large retention. The model is conditioned on the synthetic rows and scored on real holdout rows. Mitra and TabICL must be measured on the same mapped tables for the gate. Their retentions are not averaged into the bound. A CatBoost retention cannot fill this component.

Real fit rows, real holdout rows, and synthetic rows share one map. The map is fit on the real fit rows only. Numeric columns go to the open unit interval by an empirical mid-rank quantile. Categorical levels are pre-encoded to fixed codes in that same interval. Column names become HMAC digests. The map does not publish a minimum or a maximum. A second fit on the synthetic rows is a protocol failure. The tabular auditors and the embedding distance both see this mapped pair.

Representation closeness asks whether Kumo large places the mapped synthetic table near the mapped real table. The distance is half the Euclidean length between the two unit-length vectors, and it lies in $[0, 1]$. Closeness is one minus that distance divided by the distance to a matched null. The synthetic table must also sit closer than an independent-marginal table of the same size, under the same map. The matched null is a second table whose cheap-model scores sit in a band fixed before the synthetic rows are encoded. Mitra and TabICL must show the same sign as Kumo large on that gap. A sign flip, a zero match distance, or a missing vector makes the component null. The component is required. It does not stand in for the tabular transfer.

The headline encoder is Kumo large. Its loss is the structured-data-model regressor's mean of the inverse-transformed quantile forecasts. A single in-context prediction is a different scale and is not this loss. Mitra and TabICL repeat that transfer on the same mapped tables. Both are required for the tabular gate. Neither is averaged into representation closeness. Mitra is the zero-shot regressor. TabICL restores the context scale before the squared error. Kumo small and the project encoder a0 were not run. Kumo is available under the OpenMDW license and the Apache license. Mitra is Apache-licensed. TabICL is BSD-licensed. The chase score refuses Foundation and TabPFN, whose licenses are not in the clean set for this release. It also refuses the target encoder, the historical context assembly, Hyperion, TabDPT, and the target-aware blocks. TabDPT's weights are Apache-licensed and still refused, because that model was trained on real tables that overlap this panel. Hyperion is the project's and still refused, because one training arm saw labels. The 4,168-dimensional vector remains the router embedding. It is not an input to (3). The shipped generator remains `features12.steps2048`.

The object a recipient keeps is that generator together with a hash manifest: a salt identifier, the normalizer name, the encoder pin, one digest per column, a dtype code, and the byte length. The manifest carries no row, header, category string, extremum, or encoder weight. The salt and the hash-to-label map stay in a local file the recipient holds. A public sidecar that echoes a source string

or an extremum fails the cleartext gate. An exact copy of a real row fails its own gate. Crossing 10,240 bytes fails the byte gate. In each case the scalar stays null.

The tabular error and the embedding distance answer different questions. Tabular transfer asks whether a clean tabular model conditioned on the synthetic rows recovers the improvement that the same model obtained from real fit rows. Representation closeness asks whether the mapped synthetic table sits near the mapped real table. A release needs both measurements. A copied row, a cleartext string, or a byte-cap failure still removes the score.

PTF-v1 is the other release KPI. Its hard gates include joint fidelity, membership and attribute attacks, copy checks, and feature-importance agreement on a sealed holdout. It is null here, as are MFS-v2 and MFS-v3. Compile-time search records the proxy $0.45u + 0.25d + 0.30j$ and then keeps the candidate with the smallest gate shortfall, then the smallest artifact. That record withholds a release certificate. The proxy is not (3).

No artifact in this paper has a measured MFS-v2 or MFS-v3 value. Emitting MFS-v2 requires the six components, a non-empty hard-gate map that passed, both the size- n and size- $4n$ artifacts, real-versus-real controls, artifact-only sampling, and locked metric implementations. Emitting MFS-v3 also requires the clean tabular transfer and the embedding distance under the shared map, and every hard gate has to pass. Table 5 reports the measured retentions. A retention is not the scalar. The cleartext scan is unscanned where the stored column codes are internal identifiers, and other gates fail on many lineages, so the scalar stays null. A retention win does not supply those inputs. Copying a retention number into (3) would publish a value the contract refuses to emit. The supplement applies (3) to invented components, including a zero compactness term. Those rows are arithmetic. They show the shape of the product. They are not a score for any method in the tables. Baselines stay on their native objectives. The shared columns in this paper stay retention and charged bytes.

3 EXPERIMENTAL SETUP

3.1 PARTITIONS AND PROVENANCE

The panel is the rights-cleared PMLB regression collection (Olson et al., 2017; Romano et al., 2022). Of these prepared lineages, 20 names begin with feynman, which the PMLB catalog describes as synthetic physics, 14 begin with strogatz, which that catalog describes as simulated dynamics, 25 contain the Friedman-style fri marker, 6 contain bng and are Bayesian-network generators, and 35 are the remainder. This paper does not re-verify which of the remainder are observational measurements. Distinct lineage identifiers are not a claim of independent source families. Related names, including the cpu_small lineages, stay inside the one-row-per-lineage bootstrap. The resampling unit is the lineage, so related variants move together only by chance. A source-family interval is not published, because a verified parent map is not in the ledger. The data lock admits 104 MIT-licensed entries and prepares 100. 4 lineages have no split object because the same source-row digest appears in two partitions (check_nonoverlap raises on that overlap). Another 3,755 catalog entries lack a redistribution grant and are not backfilled. The prepared cut is a grouped 80/20 of the official training rows, seed 1729. Worker directories hold train.csv and validation.csv. They do not hold test.csv. Official tests stay sealed, and final evaluation is not authorized. The fit view has 72 to 800 rows. Sources above the quota use a hybrid coreset that keeps the observed feature and target range. The supplement records that draw. Held-out validation rows are the grouped holdout of the official training file, and those source rows include the simulated generators named above. Table 1 is the lock and the file-name count. Test-file contents were not read.

Fit seed 11 defines the primary displayed DOPE and legacy comparison populations. Later retained panels state their own fit-seed coverage. Sizes n and $4n$ are both stored. The tables below are $4n$. The Gaussian copula row is Copulas 0.14.1 GaussianMultivariate, selected by held-out mean log density. CTGAN and TVAE are the author implementation, selected by the mean of linear and MLP training-mean skill on validation rows, the same skill defined in Section 4. Chow-Liu uses eight bins and Laplace smoothing 1.0. It is an implementation written for this comparison, as are the independent marginals. The ARF watch log's last count is 799 closed and 799 ok out of 800 planned, then a sealing record. The later matched ARF publication supplies scored retention; its displayed-profile comparison median is 0.840. The new retained diagnostic panel has a separate

Table 1: **Test-split provenance.** File-name counts only.

Item	Record
Catalog SHA-256	ab9fda8d2dea46067b70a42812e9d3c1d9dc6ba025780100df34377e81aa1120
Eligible / prepared / overlap exclusions	104 / 100 / 4
Unknown-rights catalog entries held out	3755
Split	official_test.grouped_training_80_20, seed 1729
Overlap lineage ids	590210721973399a, 9fe58a656fdb458a, d06a584f07d0acd5, d0ba629ed07b5d63
Final evaluation authorized	false
S3 worker directories / train.csv / validation.csv	100 / 100 / 100
test.csv under S3 workers	0
BeyondArena workers / worker test.csv	5 / 0
Evaluator test.csv files, contents not read	5

Table 2: **Fit time, memory, GPU, and charged bytes where a fit timer was stored.**

Method	Fit s	RAM / VRAM	Hardware	GPU h	Bytes
DOPE	7.25(13.5)	920.7 / 1067	one NVIDIA GeForce RTX 4090 and two NVIDIA GeForce RTX 3090 GPUs	0.193	1,895.5(758–22,233)
CTGAN	213.5(268.1)	1557.8 / 1101	NVIDIA GeForce RTX 4090	2.52	741,834(528,075–1,428,695)
TVAE	214.2(253.2)	1252.9 / 1069	NVIDIA GeForce RTX 4090	2.45	317,141.5(157,578–741,659)
Forest-Flow	185.9(380.6)	6112.0 / 3181	NVIDIA GeForce RTX 4090	2.44	325,848,913.5(115,202,253–1,085,155,062)

receipt and aggregation rule. Forest-Flow on 6 lineages uses the author regression objective on its frozen grid. Native objectives are not ranked across methods.

3.2 STATISTICS

Comparisons are paired on the lineages where both methods have a complete informative group. The tests are two-sided Wilcoxon signed-rank tests on the nonzero paired differences, and one-sided sign tests for a positive difference, with a matched-pairs rank-biserial correlation. Holm correction is applied inside a family, separately for each test, and is not pooled across families. On CatBoost the Holm sign-test values are 2.15×10^{-15} , 6.72×10^{-24} , and 4.95×10^{-27} . The density family has nine tests (three comparators times three auditors). The neural family is CTGAN and TVAE times the auditors. The Forest-Flow family is separate and has six lineages. A Friedman test and a Nemenyi critical difference use the four density methods on their common complete-informative set, with rank 1 assigned to the higher retention. The critical-difference constant for four methods at $\alpha = 0.05$ is 2.569. The primary comparison interval resamples informative lineages. Each lineage contributes the median of sample seeds 101, 211, and 307 from the single fit at seed 11. Later ARF and Forest-Flow panels report between-fit variation separately. This primary interval does not include a new fit, a new split, a new population, or a change of auditor. These are descriptive validation tests. They are not the release KPI in Section 1.2, and they do not authorize a release at the 10,240-byte tier. MFS-v2 and PTF-v1 stay null.

3.3 COMPUTE

Table 2 lists fit time, peak memory, the GPU models, GPU hours, and charged bytes for receipts that store a fit timer. Sample-timer medians (maxima) are 186.7 (197.0) for CTGAN, 186.1 (196.6) for TVAE, and 342.4 (511.7) for Forest-Flow. The DOPE receipt stores no sample timer. No receipt stores a separate CPU timer. Gaussian copula, Chow–Liu, independent marginals, and ARF have no fit timer, so they are omitted; their byte medians are in Table 7. ARF receipts do not record a GPU model. TabSyn and TabDDPM costs are not measured and not measured. The supplement states the replay walls, the Forest-Flow sums, and the unsplit density attempt timers.

4 RESULTS

Strongest tuned baseline. Table 3 keeps, for each block and auditor, the tuned baseline with the highest paired median retention. The difference is DOPE minus that baseline. Its interval is the paired lineage bootstrap already used in the block tables. W/T/L counts lineages with a positive, zero, or negative difference. Holm p is the adjustment inside that block’s existing family. It is not a new family, and the blocks are not pooled. A positive median with a small p -value still includes

Table 3: **Strongest tuned baseline at $4n$.** W/T/L counts lineages. Holm stays inside the block.

Block	Auditor	Baseline	n	Difference	W/T/L	Holm p
Density	CatBoost	Gaussian copula	97	0.224 [0.149, 0.290]	86/0/11	1.35×10^{-13}
Density	Linear	Gaussian copula	92	0.00366 [-0.00438, 0.012]	49/0/43	0.373
Density	MLP	Gaussian copula	75	0.185 [0.128, 0.228]	65/0/10	4.96×10^{-11}
Neural	CatBoost	TVAE	21	0.236 [0.087, 0.431]	19/0/2	1.57×10^{-4}
Neural	Linear	TVAE	21	0.263 [0.166, 0.410]	20/0/1	1.57×10^{-4}
Neural	MLP	TVAE	19	0.284 [0.153, 0.734]	17/0/2	1.57×10^{-4}
Forest	CatBoost	Forest-Flow	6	-0.027 [-0.308, 0.00804]	1/0/5	0.375
Forest	Linear	Forest-Flow	5	0.012 [-0.260, 0.302]	3/0/2	0.812
Forest	MLP	Forest-Flow	4	-0.032 [-0.227, -0.010]	0/0/4	0.375

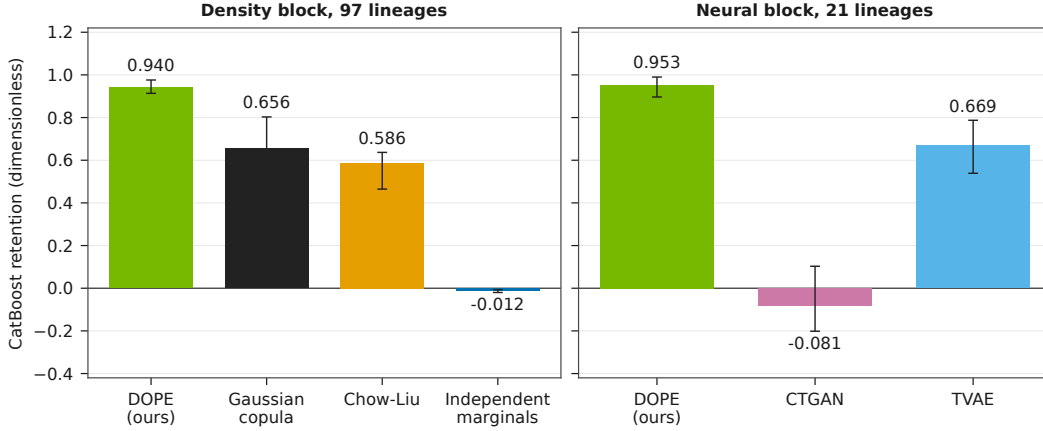


Figure 3: **CatBoost retention at $4n$.** Higher is better. Each bar is the paired-lineage median and its lineage-bootstrap interval. Density uses 97 lineages. The neural panel uses 21 and is not pooled. Forest-Flow, 6 lineages, is higher on its own block and is omitted.

lineage losses. The paired median difference is not the difference of the two medians. 0.224 is not 0.940 minus 0.656.

CatBoost retention. Fig. 3 shows the level of that shared utility under CatBoost. Retention is the fraction of a real-trained auditor’s improvement over a null that an auditor trained on synthetic rows recovers. Higher is better. The value is not MFS-v2. MFS-v2 stays null on every method. Each bar is the median on the paired lineages in that block, with the lineage-bootstrap interval. The two panels are not pooled. Forest-Flow is a separate 6-lineage block. Its CatBoost median is higher than the paired DOPE median on that block, the Holm test is not significant, and that block is not drawn here.

4.1 DENSITY BLOCK

Density marks. Circles are DOPE, squares the Gaussian copula, triangles Chow–Liu, and diamonds the independent marginals. Filled marks are lineages. Open marks are that method’s median and its lineage-bootstrap interval. Table 4 is the paired density block. Under CatBoost, DOPE’s median paired differences are positive against the Gaussian copula (0.224, interval 0.149 to 0.290), the Chow–Liu tree, and the independent marginals. The Holm-adjusted p -values are 1.35×10^{-13} , 4.85×10^{-16} , and 1.17×10^{-16} . The rank-biserial correlations are 0.890, 0.978, and 0.999. The Gaussian copula is the strongest density baseline. Wins, ties, and losses against it are 86/0/11 under CatBoost and 65/0/10 under the MLP. On the 97 lineages common to all four methods, the Friedman p -value is 2.71×10^{-44} and the critical difference is 0.476. All six CatBoost pairs exceed that difference. Fig. 4 shows the sorted paired differences with the same four marks. The horizontal axis is the retention difference. The segment is the critical difference.

Table 4: **Density block at $4n$.** Each cell is a dimensionless retention. W/T/L counts lineages. Holm p

Auditor	Comparator	n	DOPE	Other	Difference	W/T/L	Holm p
CatBoost	Gaussian copula	97	0.940 [0.914, 0.976]	0.656 [0.431, 0.803]	0.224 [0.149, 0.290]	86/0/11	1.35×10^{-13}
CatBoost	Chow-Liu	97	0.940 [0.914, 0.976]	0.586 [0.465, 0.637]	0.337 [0.282, 0.451]	94/0/3	4.85×10^{-16}
CatBoost	Indep. marginals	97	0.940 [0.914, 0.976]	-0.012 [-0.020, -0.00649]	0.956 [0.918, 0.980]	96/0/1	1.17×10^{-16}
Linear	Gaussian copula	92	0.994 [0.981, 0.996]	0.989 [0.968, 0.994]	0.00366 [-0.00438, 0.012]	49/0/43	0.373
Linear	Chow-Liu	92	0.994 [0.981, 0.996]	0.661 [0.356, 0.738]	0.340 [0.244, 0.425]	86/0/6	7.12×10^{-15}
Linear	Indep. marginals	92	0.994 [0.981, 0.996]	0.00206 [-0.023, 0.00583]	0.988 [0.975, 1.001]	92/0/0	5.70×10^{-16}
MLP	Gaussian copula	75	1.061 [1.024, 1.125]	0.928 [0.865, 0.953]	0.185 [0.128, 0.228]	65/0/10	4.96×10^{-11}
MLP	Chow-Liu	75	1.061 [1.024, 1.125]	0.692 [0.582, 0.754]	0.530 [0.350, 0.651]	74/0/1	2.11×10^{-13}
MLP	Indep. marginals	75	1.061 [1.024, 1.125]	-0.028 [-0.047, -0.013]	1.104 [1.047, 1.179]	75/0/0	2.11×10^{-13}

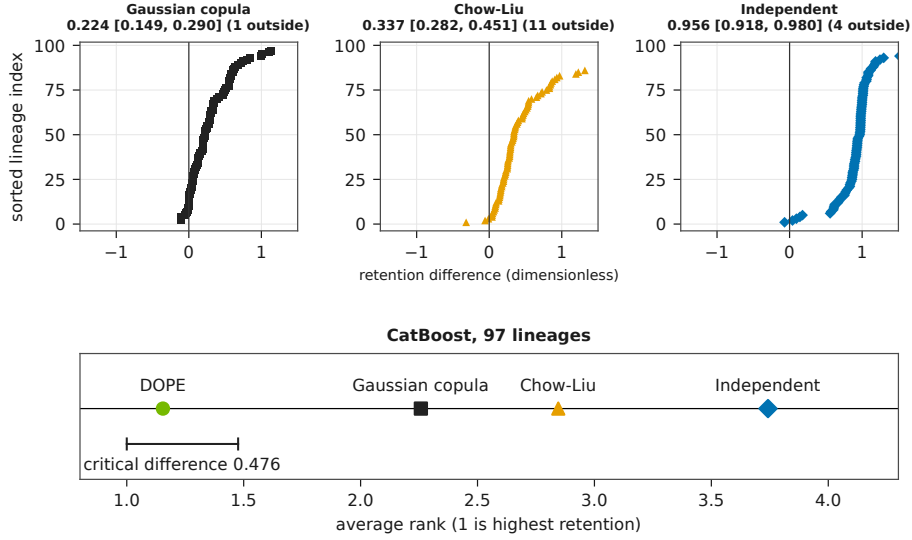


Figure 4: **CatBoost paired differences, sorted, and the Nemenyi scale.** Rank 1 is the higher retention. The segment length is the critical difference.

Retention against charged bytes. Fig. 5 plots the same density-block retention against $\log_{10}$ charged bytes. The table intervals are the paired subsets. For CatBoost those two DOPE sets coincide. Paired differences outside $[-1.5, 1.5]$ are counted in the panel title and omitted from that frame. The retention axis stops at 1.6. The dashed line is the 10,240-byte release gate. Crossing it is not a retention of zero, and a retention above the line is not a release. Neural and Forest-Flow artifacts are not on this figure.

Clean-license tabular transfer. Table 5 is the same density block scored by Kumo large, Mitra, and TabICL. The shared map and the three sample seeds are the ones named above. These auditors are not pooled with the CatBoost, linear, and MLP rows. Fig. 6 shows the lineage medians. Fig. 7 shows the sorted paired differences. Squares are the Gaussian copula, triangles are Chow-Liu, and diamonds are the independent marginals.

Release scalar on this receipt. Kumo large averages 16 inverse-transformed quantile forecasts. The receipt scores 0 of 388 lineages, so the scalar is null. Cleartext is unscanned on 388 lineages.

Kumo large	row.project.output.mean.labeled.rows
Mitra	final.layer.norm.input.mean.self.query
TabICL	tficl.output.mean.self.query

Linear non-separation. The linear auditor does not separate DOPE from the Gaussian copula (49/0/43; Holm 0.373; difference 0.00366, interval -0.00438 to 0.012 ; rank-biserial 0.107). The interval contains zero. That non-separation is not an equivalence test. Ranks are 1.533 and 1.598 against critical difference 0.489. Chow-Liu and the independent marginals remain separated under the linear auditor and the MLP. A CatBoost separation is not a separation under every auditor.

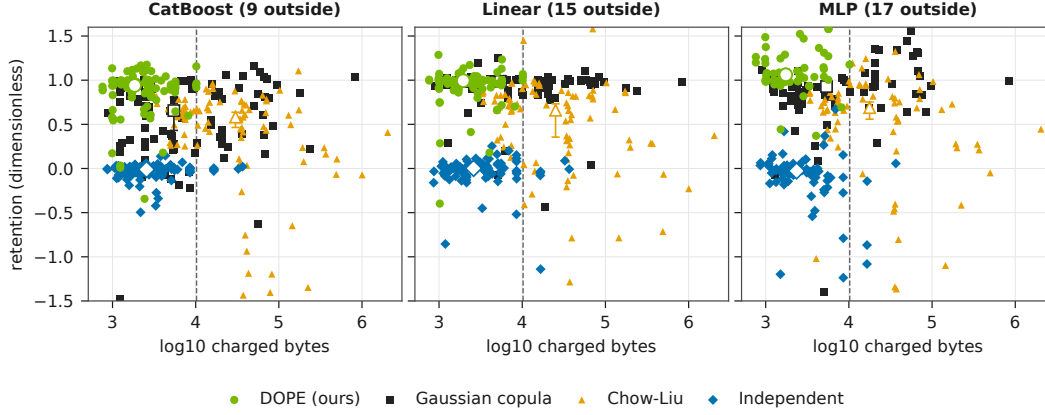


Figure 5: **Density-block retention at $4n$ against charged bytes.** The dashed line is the release byte gate.

Table 5: **Kumo large, Mitra, and TabICL on the density block at $4n$.** Each cell is a dimensionless retention, not an MFS value. W/T/L counts lineages.

Auditor	Comparator	n	DOPE	Other	Difference	W/T/L	Holm p
Kumo large	Gaussian copula	97	0.863 [0.809, 0.880]	0.747 [0.668, 0.778]	0.059 [0.0098, 0.116]	62/0/35	5.70×10^{-4}
Kumo large	Chow-Liu	97	0.863 [0.809, 0.880]	0.562 [0.501, 0.629]	0.247 [0.195, 0.306]	85/0/12	1.34×10^{-11}
Kumo large	Indep. marginals	97	0.863 [0.809, 0.880]	-0.000819 [-0.00189, 7.58e-05]	0.874 [0.834, 0.888]	94/0/3	7.46×10^{-17}
Mitra	Gaussian copula	97	0.858 [0.803, 0.887]	0.713 [0.585, 0.771]	0.066 [0.023, 0.135]	65/0/32	1.34×10^{-4}
Mitra	Chow-Liu	97	0.858 [0.803, 0.887]	0.583 [0.527, 0.643]	0.242 [0.182, 0.293]	85/0/12	2.71×10^{-11}
Mitra	Indep. marginals	97	0.858 [0.803, 0.887]	-0.00907 [-0.014, -0.00377]	0.890 [0.840, 0.909]	94/0/3	6.01×10^{-17}
TabICL	Gaussian copula	97	0.871 [0.835, 0.893]	0.754 [0.672, 0.787]	0.079 [0.00947, 0.116]	62/0/35	1.61×10^{-4}
TabICL	Chow-Liu	97	0.871 [0.835, 0.893]	0.587 [0.484, 0.650]	0.267 [0.200, 0.302]	87/0/10	7.12×10^{-12}
TabICL	Indep. marginals	97	0.871 [0.835, 0.893]	-0.00446 [-0.012, 0.0012]	0.891 [0.846, 0.919]	92/0/5	5.83×10^{-17}

Raw auditor loss. Table 6 is the median raw loss on the same informative lineages. RMSE is the square root of the stored mean squared error. Training-mean skill is one minus that loss divided by the training-mean null loss. Conventional held-out R^2 uses the mean of the evaluation targets and was not stored for these rows. Retention is the median of the per-lineage ratios, not the ratio of these medians. CatBoost reads RMSE 0.053 and 0.081, with training-mean skill 0.881 and 0.792.

Charged bytes. Table 7 counts artifacts at or under the cap. The two displayed DOPE artifacts over the cap are 4544.GeographicalOriginalofMusic at 22,233 bytes and 588_fri_c4_1000_100 at 19,019 bytes. 678_visualizing_environmental has a fit whose CatBoost group is not informative, so the CatBoost median uses 97 lineages while the byte count uses 100. Independent marginals pass the cap more often than the Chow-Liu baseline and still sit near retention zero. Smallness alone is not the result. The displayed byte range runs from 758 to 22,233.

Fidelity and empirical privacy. On the density cells, the displayed profile has median classifier two-sample AUC 0.517 (interval 0.504 to 0.532, $n = 98$), mean marginal Kolmogorov-Smirnov distance 0.114 (0.103 to 0.135), and pairwise correlation fidelity 0.953 (0.945 to 0.961). A two-sample AUC near one half means a linear-time classifier has trouble telling the synthetic rows from the held-out validation rows. Table 8 adds alpha-precision, beta-recall, the unbalanced DCR share, and a DOMIAS AUC. The lower block is the neural lineages only. Holm families are 12 and 8, separate from retention. The DCR column is the fraction of synthetic queries whose nearest reference is in the fit view, with ties scored one half. A balanced variant was computed in the harness and is not this column. Under an independent draw, the unbalanced share sits near the fit-view fraction of the pooled references. The grouped split keeps about four fifths of those rows in the fit view, so the baseline is near four fifths, not one half. Higher values of that share, and higher DOMIAS AUC, sit nearer the fit view and are not a quality win. DOMIAS here is the isolated kernel density-ratio equation from the pinned author expression, evaluated on this numeric projection. It is not the original DOMIAS training pipeline and it is not a membership-inference certificate. The DOMIAS AUC stays near one half. The density DCR contrast against independent marginals clears Holm in that worse

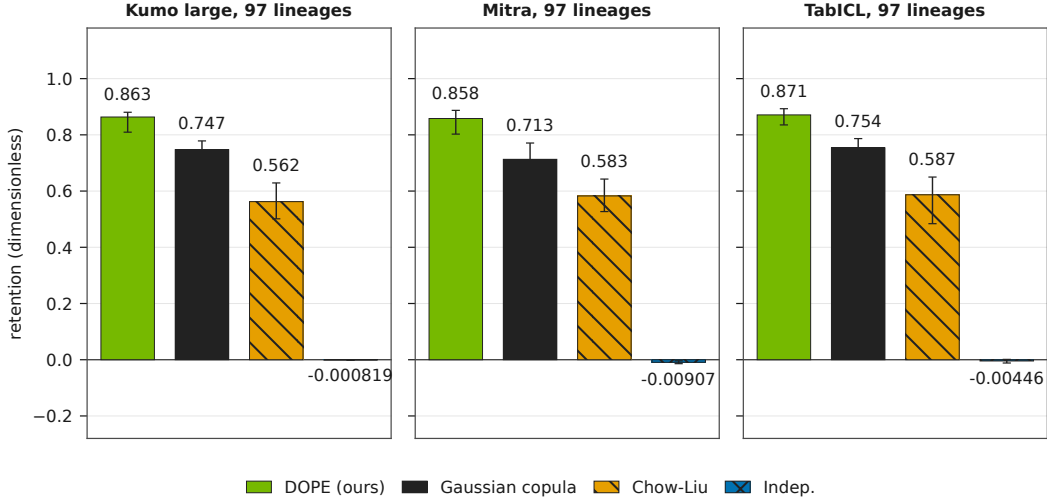


Figure 6: **Kumo large, Mitra, and TabICL retention at $4n$.** Higher is better. Each bar is the lineage median and its lineage-bootstrap interval. Hatches distinguish the methods. This panel is not an MFS value.

Table 6: **Raw auditor medians on the informative lineages.**

Auditor	n	RMSE real	RMSE synth.	training-mean skill, real	training-mean skill, synth.
CatBoost	97	0.053	0.081	0.881	0.792
Linear	92	0.105	0.112	0.585	0.551
MLP	75	0.098	0.083	0.666	0.785

direction. The displayed-profile ARF publication median is 0.840. The pre-specified privacy-attack panel remains not measured. This is not differential privacy or HIPAA de-identification.

4.2 NEURAL BLOCK

Neural block. Table 9 is not pooled with Table 4. On the 21 CatBoost lineages, DOPE’s median is 0.953, CTGAN’s is -0.081 , and TVAE’s is 0.669. TVAE is the stronger neural baseline. Wins, ties, and losses against TVAE are 19/0/2, 20/0/1, and 17/0/2. The median favors DOPE, and DOPE still loses lineages. Every successful CTGAN and TVAE artifact in this byte count is over the cap (0/22 and 0/22). Runs that stopped at the recorded time limit are omitted. They are not counted as failures of a finished fit.

4.3 FOREST-FLOW

Forest-Flow confirmation. Table 10 is a 6-lineage confirmation, not pooled with the density or neural blocks. The Forest-Flow CatBoost median is 1.003 (interval 0.926 to 1.026) and the paired DOPE median is 0.940. Wins, ties, and losses are 1/0/5 under CatBoost, 3/0/2 under the linear auditor, and 0/0/4 under the MLP. DOPE is lower on every paired MLP lineage. The Holm tests are not significant at this count, so this is not a certified win for either method. Charged Forest-Flow artifacts run from 115,202,253 to 1,085,155,062 bytes (median 325,848,913.5) against a DOPE median of 2,067 on the same lineages. MFS-v2 on this block is null.

Table 11 is the published matched-population receipt for ARF at $4n$ on the displayed profile. Complete-population TabDDPM, TabSyn, and Forest-Flow receipts are omitted; the completed retained TabSyn subset is reported separately below.

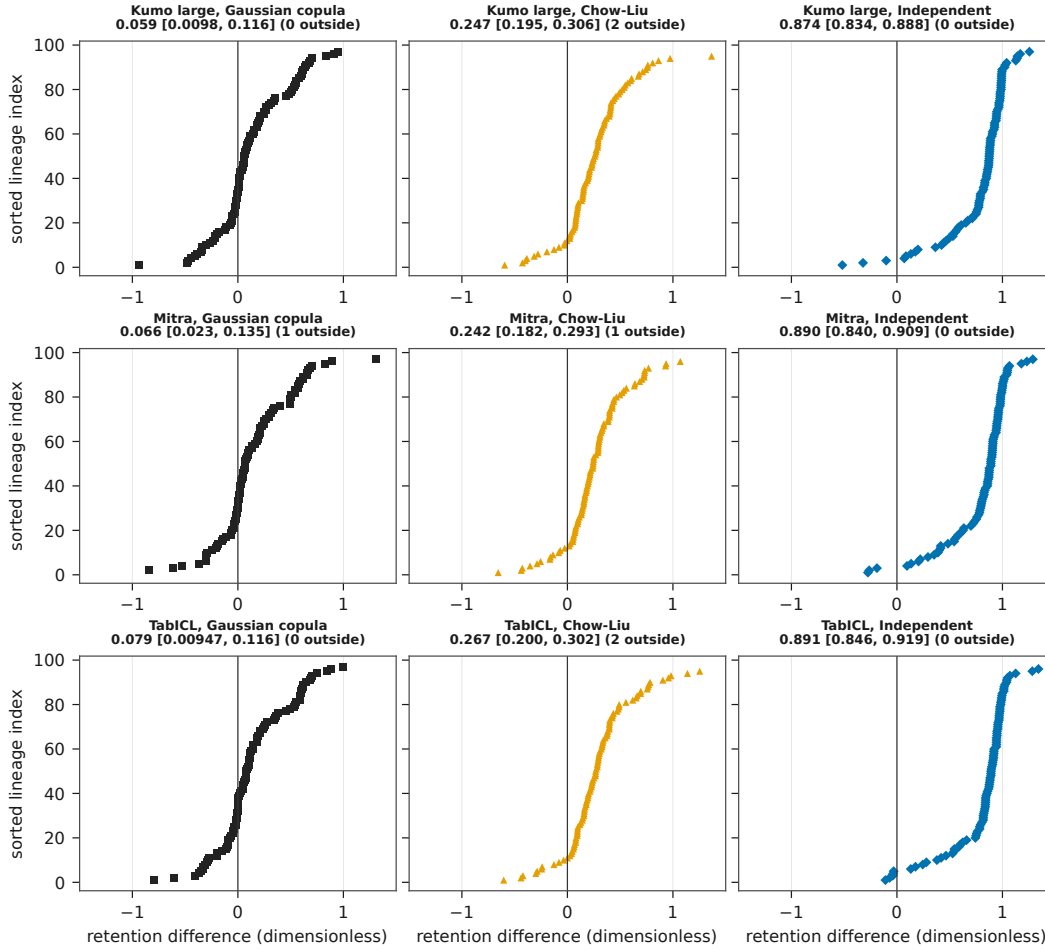


Figure 7: **Paired retention differences for Kumo large, Mitra, and TabICL, sorted by lineage.** Filled marks are lineages.

Table 7: **Charged bytes at $4n$.** Forest-Flow is not pooled.

Method	Median bytes	Within cap
DOPE reference	1,895.5	98/100
Gaussian copula	5,464.5	62/100
Chow-Liu	29,865	31/100
Independent marginals	2,549.5	94/100
CTGAN	741,834	0/22
TVAE	317,141.5	0/22
Forest-Flow	325,848,913.5	0/6
ARF	1,627,718.5	0/100

4.4 RETAINED VALIDATION EVIDENCE

Retained validation. Tables 12 and 13 add the merged retained-sample evidence. Each lineage averages all three sample seeds before a lineage-bootstrap median and percentile interval are computed. This differs from the median-of-seed rule in the earlier blocks; those blocks and their numbers are preserved. A lineage enters a metric only when all three values exist. An unavailable outcome stays absent. Table 12 covers the same prepared population for ARF, copula and Chow-Liu defaults and native-selected configurations. Table 13 includes only the complete scaled author-default Tab-Syn lineages and the classical methods on those identical inputs. It cannot establish a full-population

Table 8: **Grouped-validation fidelity and empirical privacy.** Not DP or HIPAA.

Method	α -prec.	β -recall	DCR share (unbalanced)	DOMIAS
<i>Median [interval] (lineages).</i>				
DOPE	0.905 [0.889, 0.918] (98)	0.437 [0.398, 0.473] (98)	0.811 [0.806, 0.814] (98)	0.508 [0.498, 0.520] (94)
Gaussian	0.812 [0.786, 0.832] (100)	0.430 [0.378, 0.453] (100)	0.806 [0.802, 0.811] (100)	0.513 [0.494, 0.527] (94)
Chow-Liu	0.858 [0.825, 0.904] (100)	0.359 [0.312, 0.400] (100)	0.810 [0.806, 0.816] (100)	0.512 [0.495, 0.522] (94)
Indep.	0.885 [0.868, 0.897] (100)	0.284 [0.197, 0.347] (100)	0.801 [0.797, 0.806] (100)	0.507 [0.496, 0.521] (94)
DOPE	0.902 [0.882, 0.940] (22)	0.441 [0.381, 0.489] (22)	0.811 [0.802, 0.819] (22)	0.508 [0.480, 0.538] (22)
CTGAN	0.719 [0.520, 0.856] (22)	0.170 [0.142, 0.267] (22)	0.820 [0.800, 0.842] (22)	0.502 [0.463, 0.534] (22)
TVAE	0.850 [0.731, 0.896] (22)	0.356 [0.283, 0.401] (22)	0.825 [0.813, 0.843] (22)	0.527 [0.487, 0.563] (22)

Contrast	α -prec.	β -recall	DCR share (unbalanced)	DOMIAS
<i>Line 2: Holm p; rank-biserial r; n. DOPE minus comparator.</i>				
	0.083 [0.045, 0.113]	0.024 [0.011, 0.039]	-0.0025 [-0.00625, 0.005]	0.00157 [-0.0069, 0.0074]
DOPE–Gaussian	1.86×10^{-10} ; 0.781; 98 0.019 [0.00782, 0.056]	0.394; 0.215; 98 0.048 [0.029, 0.072]	1.000; 0.000206; 98 -0.00125 [-0.00562, 0.00125]	1.000; -0.085; 94 0.000581 [-0.00802, 0.011]
DOPE–Chow-Liu	2.00×10^{-5} ; 0.551; 98 0.018 [0.0072, 0.028]	1.32×10^{-10} ; 0.789; 98 0.104 [0.068, 0.138]	0.806; -0.169; 98 0.00625 [0.0025, 0.011]	1.000; 0.00572; 94 0.00244 [-0.00595, 0.015]
DOPE–Indep.	5.47×10^{-5} ; 0.523; 98	9.71×10^{-15} ; 0.937; 98	0.005; 0.398; 98	1.000; 0.080; 94
DOPE–CTGAN	0.168 [0.054, 0.368] 0.004; 0.794; 22	0.219 [0.131, 0.256] 3.81×10^{-6} ; 1.000; 22	-0.00812 [-0.029, 0.015] 0.532; -0.277; 22	-0.00028 [-0.014, 0.031] 0.610; 0.130; 22
DOPE–TVAE	0.071 [0.00713, 0.141] 0.040; 0.644; 22	0.062 [0.00546, 0.135] 0.069; 0.573; 22	-0.025 [-0.030, -0.00641] 0.059; -0.613; 22	-0.021 [-0.036, -0.00972] 0.108; -0.510; 22

Table 9: **Neural block at $4n$, not pooled.** W/T/L counts lineages.

Auditor	Comparator	n	DOPE	Other	Difference	W/T/L	Holm p
CatBoost	CTGAN	21	0.953 [0.896, 0.990]	-0.081 [-0.202, 0.103]	0.881 [0.702, 1.150]	21/0/0	5.72×10^{-6}
CatBoost	TVAE	21	0.953 [0.896, 0.990]	0.669 [0.539, 0.787]	0.236 [0.087, 0.431]	19/0/2	1.57×10^{-4}
Linear	CTGAN	21	0.999 [0.996, 1.015]	-0.033 [-0.225, 0.080]	1.049 [0.916, 1.236]	21/0/0	5.72×10^{-6}
Linear	TVAE	21	0.999 [0.996, 1.015]	0.796 [0.605, 0.867]	0.263 [0.166, 0.410]	20/0/1	1.57×10^{-4}
MLP	CTGAN	19	1.026 [0.998, 1.263]	-0.149 [-0.471, 0.083]	1.079 [0.877, 2.030]	19/0/0	1.53×10^{-5}
MLP	TVAE	19	1.026 [0.998, 1.263]	0.753 [0.478, 0.852]	0.284 [0.153, 0.734]	17/0/2	1.57×10^{-4}

TabSyn result or a native-tuned TabSyn comparison. These evaluations did not select a configuration.

Table 14 gives exploratory TabSyn-minus-comparator retention differences on shared lineages. The Holm family comprises four metrics (CatBoost retention, mean marginal KS/TV, CatBoost C2ST AUC and distance-MIA AUC) times six classical configurations. Conservative binomial/order-statistic intervals for the median difference use Bonferroni correction over the entire family, conditional on independent lineages. This small cohort requires unbounded simultaneous intervals. Finite tail-adjusted bootstrap ranges are retained only as exploratory descriptions; they do not guarantee familywise coverage. Marginal intervals in Fig. 8 are not simultaneous intervals. The machine-readable reduction includes every diagnostic contrast and all applicability exclusions. A raw AUC difference is not a directional utility or privacy win. Selection of the completed subset, one fit seed and its small lineage count limit this exploratory analysis.

Table 15 separately extends the published DOPE–ARF retention comparison with lineage intervals. It preserves that receipt’s median-of-three rule. Its six-test family contains two ARF configurations and three auditors. The shared-lineage intersections differ by auditor. These validation descriptions leave the release and superiority gates null.

The supplement prints the deterministic first-batch diagnostics per physical job. Incomplete seed groups and mixed research profiles prevent population confidence intervals or method rankings for that batch. All reductions trace to committed receipt paths, source digests and JSON pointers in docs/whitepaper/generated/retained-evidence.json. The original receipt references are retained in research/benchmark/results/retained-classical-validation/panel.json and the first-batch publication recorded in that reduction.

Table 10: **Forest-Flow confirmation at $4n$, CatBoost.** Not pooled.

Lineage	DOPE bytes	DOPE	Forest-Flow bytes	Forest-Flow
1193_BNG_lowbwt	2,436	0.970	416,921,273	1.004
503_wind	3,846	0.996	637,056,689	1.016
582_fri_c1_500_25	5,670	0.909	1,085,155,062	0.883
596_fri_c2_250_5	1,698	0.807	234,776,554	0.968
feynman_I.12.1	1,032	0.991	123,784,681	1.001
strogatz_vdp1	984	0.581	115,202,253	1.036

Table 11: **Published ARF matched receipts at $4n$, CatBoost.**

Publication	Lineages	DOPE	Receipt	Difference
ARF, author_default	97	0.940	0.840	0.077
ARF, native_selected	97	0.940	0.826	0.095

The coverage table distinguishes Forest-Flow confirmation from the Forest-Diffusion pilot modes. No full-population diffusion result is inferred from the Flow subset. Additional native-scored fits and published common-metric cells are counted separately. Pending publication does not mean that a method failed. Receipt paths and digests are recorded in `docs/whitepaper/generated/baseline-coverage.json`. The ARF follow-on now supplies complete five-fit common measurements on the same validation lineages. Its between-fit standard deviation is descriptive, not a confidence interval. TabSyn’s additional retained n -sized evaluations include incomplete sample groups; those groups have no aggregate figure value. The supplementary descriptive figure inputs in `docs/whitepaper/generated/published-baseline-kpis.csv` retain a receipt path and JSON pointer for every value. Cohorts and fit counts differ; these inputs do not support a paired ranking. The ARF validation completion does not admit final official-test certification.

4.5 RELEASE SCORE

Release scalars stay null. The scalars in (3), under the pinned v2 weights and under the pre-registered v3 weights, are null for every method in this paper. Table 5 measures the clean-license tabular transfer, and the same receipt records the embedding distance. Failed gates, including an unscanned cleartext check, keep every scalar null. The supplement shows the invented v2 arithmetic. The evaluation locks for a final test were not frozen.

4.6 BEYONDARENA

BeyondArena (`2ecfe882ccfb814fc27c4de10a64ceefd5d7655c`) is not one of the 100 PMLB lineages and is not pooled with them.

5 of 12 families fit inside the cap. 7 did not prepare (4 outside the projected range, 3 overlapping). The range was not raised. Table 17 has no retention column. The supplement names all 142 families.

5 ABLATIONS

Step budget. `features12.steps8192` is the same residual with four times as many AdamW steps. It is a budget check, not the displayed model. Where both budgets have a complete three-seed group, the CatBoost median paired difference is 0.034 (68 lineages). The supplement gives the linear and MLP differences and lists all 100 lineages. A blank cell is an unavailable fit or an auditor that is not informative. It is not a zero. The scored fits did not keep a per-step series. The supplement plots the logged replay. It is not a substitute for the scored artifacts. The lower loss at 8,192 steps does not replace the displayed profile.

Table 12: **Retained classical validation at $4n$.** Median [95% lineage-bootstrap interval] (informative lineages). Native selection uses each method’s own objective.

Method	Config.	CB retention	KS/TV	C2ST CB AUC	Distance MIA AUC
ARF	Default	0.820 [0.787, 0.872] (99)	0.103 [0.097, 0.114] (100)	0.604 [0.582, 0.626] (100)	0.553 [0.538, 0.564] (100)
ARF	Native	0.806 [0.736, 0.855] (99)	0.105 [0.098, 0.114] (100)	0.610 [0.591, 0.627] (100)	0.542 [0.536, 0.559] (100)
Chow–Liu	Default	0.643 [0.552, 0.672] (99)	0.128 [0.106, 0.154] (100)	0.721 [0.652, 0.773] (100)	0.513 [0.503, 0.525] (100)
Chow–Liu	Native	0.566 [0.460, 0.631] (99)	0.126 [0.105, 0.157] (100)	0.702 [0.641, 0.761] (99)	0.513 [0.504, 0.523] (100)
Copula	Default	0.594 [0.427, 0.762] (99)	0.126 [0.094, 0.138] (100)	0.615 [0.591, 0.671] (100)	0.514 [0.503, 0.524] (100)
Copula	Native	0.641 [0.431, 0.803] (99)	0.134 [0.125, 0.158] (100)	0.730 [0.696, 0.779] (99)	0.507 [0.494, 0.521] (100)

Table 13: **Matched retained TabSyn subset at $4n$.** Same inputs and real-data controls for every method; median [95% interval] (lineages). Separate from the full classical population.

Method	Config.	CB retention	KS/TV	C2ST CB AUC	Distance MIA AUC
ARF	Default	0.839 [0.668, 0.942] (8)	0.129 [0.084, 0.165] (8)	0.623 [0.547, 0.969] (8)	0.555 [0.506, 0.613] (8)
ARF	Native	0.752 [0.534, 0.946] (8)	0.132 [0.084, 0.163] (8)	0.606 [0.583, 0.812] (8)	0.527 [0.482, 0.558] (8)
Chow–Liu	Default	0.746 [0.666, 0.859] (8)	0.186 [0.068, 0.255] (8)	0.756 [0.525, 0.999] (8)	0.522 [0.465, 0.542] (8)
Chow–Liu	Native	0.655 [0.561, 0.778] (8)	0.188 [0.087, 0.248] (8)	0.707 [0.525, 0.999] (8)	0.499 [0.470, 0.533] (8)
Copula	Default	0.643 [0.270, 0.935] (8)	0.146 [0.067, 0.242] (8)	0.687 [0.554, 0.987] (8)	0.510 [0.466, 0.541] (8)
Copula	Native	0.680 [0.260, 0.858] (8)	0.157 [0.097, 0.242] (8)	0.698 [0.642, 0.987] (8)	0.493 [0.463, 0.525] (8)
TabSyn	Scaled default	0.911 [0.685, 0.971] (8)	0.136 [0.120, 0.157] (8)	0.670 [0.597, 0.869] (8)	0.571 [0.547, 0.619] (8)

The primary informativeness threshold is 0.01 times the absolute null loss. Table 18 lists the CatBoost median by threshold and the informative counts. The middle row is the 0.01 rule behind the retention tables. Thresholds 0, 0.001, and 0.05 keep that median and that count. At 0.1 the count is 96 and the median is 0.942. Fit-seed variance across seeds other than 11 is not measured for the displayed DOPE generator.

6 RELATED WORK

Tabular synthesis has moved from parametric copulas (Patki et al., 2016) and Chow–Liu trees (Chow & Liu, 1968) through conditional GANs, including TableGAN, and variational autoencoders (Xu et al., 2019; Park et al., 2018) to diffusion and score-based models (Kotelnikov et al., 2023; Kim et al., 2023; Lee et al., 2023; Zhang et al., 2024; Shi et al., 2025; Mueller et al., 2025; Jolicœur-Martineau et al., 2024; Zagardo, 2026). Autoregressive and language-model generators (Borisov et al., 2023; Solatorio & Dupriez, 2023), prior-data fitted generators (Ma et al., 2023; Hollmann et al., 2025), CTAB-GAN and CTAB-GAN+ (Zhao et al., 2021; 2024), and graphical neural generators (Liu et al., 2023) target the same mixed-type tables. ARF estimates density with a forest (Watson et al., 2023). CART synthesis is the synthpop line (Nowok et al., 2016). A recent survey collects these design choices (Borisov et al., 2024). This paper runs the Gaussian copula, a Chow–Liu baseline written for this comparison, independent marginals written for this comparison, CTGAN, TVAE, and a 6-lineage Forest-Flow confirmation. New retained panels also measure ARF and a scaled author-default TabSyn subset (Section 4.4). The other families named above are cited and not measured; Table 16 records that coverage. SMOTE (Chawla et al., 2002) was not run. This panel scores regression retention, not classification oversampling.

Differentially private Bayesian networks and marginal synthesizers (Zhang et al., 2014; McKenna et al., 2021; 2022; Cai et al., 2021) optimize a privacy budget. Membership inference and shadow-model attacks are a separate detection literature (Shokri et al., 2017; Hayes et al., 2018; van Breugel et al., 2023). Formal differential privacy is a different claim from an empirical resemblance score (Dwork & Roth, 2014; Abadi et al., 2016). Utility reviews separate marginal fidelity from downstream prediction (Snok et al., 2018; Raab et al., 2018). Alpha-precision, beta-recall, SDMetrics, and synthcity are the usual fidelity reports (Alaa et al., 2022; Zhang et al., 2020; Qian et al., 2023). The fidelity results add alpha-precision, beta-recall, a distance-to-closest-record share, and a DOMIAS AUC. ARF and the retained TabSyn subset have an additional fidelity, detection and empirical-privacy panel in Section 4.4. The displayed-profile ARF publication median is 0.840. The pre-specified privacy-attack panel remains not measured. Nothing in those fields is a formal privacy guarantee or a HIPAA de-identification claim.

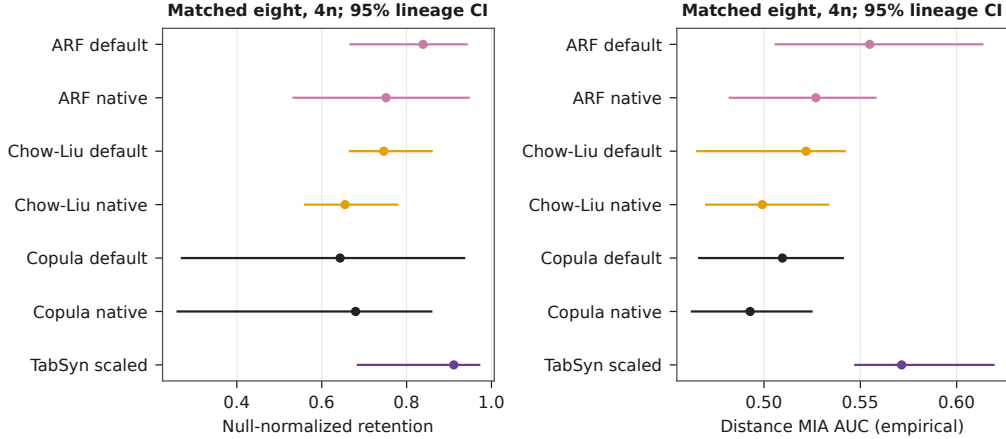


Figure 8: **Matched retained subset: median and marginal lineage-bootstrap intervals.** Privacy AUC is an empirical diagnostic; its interval is not a privacy guarantee.

Table 14: **Exploratory matched-subset CatBoost differences.** Conservative simultaneous median intervals and Holm p over the entire four-metric family; the corrected intervals are unbounded.

TabSyn minus	n	Median CB difference	Simultaneous median CI	Holm p
ARF/default	8	0.010	Unbounded	1.000
ARF/native	8	0.159	Unbounded	0.469
Chow-Liu/default	8	0.157	Unbounded	1.000
Chow-Liu/native	8	0.149	Unbounded	0.188
Copula/default	8	0.242	Unbounded	1.000
Copula/native	8	0.245	Unbounded	1.000

Tree models remain strong tabular predictors, which is why CatBoost is one of the three auditors (Grinsztajn et al., 2022; Prokhorenkova et al., 2018; Schwartz-Ziv & Armon, 2022). TabZilla and TabArena show how brittle a single split can be (McElfresh et al., 2023; Erickson et al., 2026). The regression lineages come from PMLB (Olson et al., 2017; Romano et al., 2022). Comparing several methods on the same lineages needs a rank test and a critical difference, or a Bayesian signed-rank alternative (Demšar, 2006; Benavoli et al., 2017). This paper reports the frequentist procedure: a Friedman test and a Nemenyi critical difference on the common set, plus paired Wilcoxon tests. The Bayesian signed-rank procedure is not computed.

7 LIMITATIONS

Internal. The primary displayed DOPE and legacy comparison fits use the coreset in Section 3 and fit seed 11, and a training-derived validation cut of the official training rows. The scored retention panel is regression only. Other tasks are not a result of these tables. The supplement records the draw. The official test was not an input to training, selection, or the numbers in this paper. The informativeness rule drops a negligible real-versus-null gap. Thresholds 0, 0.001, and 0.05 keep that median and that count. At 0.1 the count is 96 and the median is 0.942. Incomplete groups are omitted, not zeroed. Chow-Liu and independent marginals are implementations written for this comparison. CTGAN and TVAE use the lineages that finished. Runs that stopped at the recorded time limit are omitted. They are not counted as failures of a finished fit.

The supplementary budget protocol retains the frozen 600-second Forest-Flow fit cap and all timed-out attempt receipts. Timeout cells have missing quality metrics and confer no DOPE win; coverage denominators include their recorded dispositions. The October 8 disposition addendum was written after 75 observed timeouts and is not retroactive preregistration. A longer-budget sensitivity track would require a separate prospective freeze and comparable method budgets, with both tracks' costs

Table 15: **DOPE–ARF legacy matched receipt at $4n$: exploratory paired differences, conservative simultaneous median intervals and Holm p .** Median-of-seed aggregation.

ARF config.	Auditor	n	DOPE minus ARF	Simultaneous median CI	Holm p
Default	catboost	97	0.077	[0.019, 0.154]	1.31×10^{-5}
Native	catboost	97	0.095	[0.028, 0.175]	5.47×10^{-7}
Default	linear	92	0.027	[0.0056, 0.090]	8.53×10^{-5}
Native	linear	92	0.037	[0.00582, 0.113]	1.62×10^{-5}
Default	mlp	75	0.115	[0.043, 0.236]	4.56×10^{-8}
Native	mlp	75	0.143	[0.040, 0.253]	8.01×10^{-8}

Table 16: **Baseline coverage from committed publications.** Pending rows have no estimated result or completion date.

Method	Published measured coverage	Remaining campaign
TabSyn	49 scaled-default lineages, 143 measured cells at fit11: eight complete $n/4n$ lineages (48 cells), plus 95 n cells on 41 lineages (19 complete/22 partial groups). Native tuning/full population pending.	Pending
TabDDPM	12 retained lineages; 126 physical/132 logical cells; 21 models, fit seed 11. Default/native cohorts differ; native search incomplete.	Pending
Forest-Flow	13 default lineages at five fit seeds; 65 models/390 cells. Separate legacy six-lineage default/native panel; full population pending.	Pending
Forest-Diffusion	Pilot only; separate 100-lineage matched diffusion panel pending.	Pending
ARF	100 default/native lineages at five fit seeds, $n/4n$, three sample seeds: 6,000 logical cells / 5,892 physical receipts (108 aliases). Between-fit SD available; official-test certification pending.	Pending
GaussianCopula	100 default/native lineages at fit seed 11; 1200 logical cells. Five-fit common coverage pending.	Pending
Chow-Liu	100 default/native lineages at fit seed 11; 1200 logical cells. Five-fit common coverage pending.	Pending

retained. The supplement’s `BASELINE_PROTOCOL.md` links the frozen completion checkpoint and every timeout disposition.

The validation cut also supplies native tuning signals. These are post-selection validation estimates, not independent official-test estimates. Lineage intervals hold the split and fitted artifact fixed; they omit fitting uncertainty. Related simulation lineages can violate the independence assumption behind both bootstrap and order-statistic intervals. The retained-subset results therefore remain exploratory.

External. The primary displayed DOPE fits do not establish behavior on the full tables used by the diffusion papers, across five fit seeds, or on the official test. The additional ARF and Forest-Flow fit seeds do not establish DOPE fitting uncertainty. Generalization past seed 1729 is not measured for the displayed DOPE generator. Forest-Flow’s byte comparison uses 6 lineages and the hash-checked fit files. BeyondArena is a separate fit check. It is not a retention panel, and it is not pooled with the density block.

Construct. Retention is the reported utility. Both release scores in Section 2.4 stay null. The linear auditor does not separate DOPE from the Gaussian copula, and that result is not an equivalence. Shipping this artifact is not the same experiment as shipping the compact predictor itself. The panel does not contain that control, so it does not show that a recipient needs the synthetic rows rather than the predictor. Normalized-space RMSE is the loss the auditors stored. Original-unit error after inverting the training transform was not stored, and this paper does not report it. Table 8 is not a membership-inference certification. Its unbalanced DCR share and DOMIAS AUC are empirical. Empirical resemblance is not differential privacy and is not HIPAA de-identification.

Table 17: **BeyondArena families that prepared.** No retention auditor.

Family	Fit rows	2,048 B	8,192 B	Val 2,048	Val 8,192
airfoil	802	804	818	0.010	0.00448
garments	939	2,317	2,372	0.028	0.025
musk	3,738	9,273	9,350	0.146	0.090
telemonitoring	257	1,852	1,890	0.033	0.019
video	35,918	5,054	5,071	0.00247	0.00195

Table 18: **Informativeness threshold and auditor counts.**

Threshold	DOPE	Gaussian	Chow–Liu	Indep.	Threshold	CatBoost	Linear	MLP
0	0.940 (97)	0.656 (99)	0.582 (99)	-0.012 (99)	0.001	97	93	75
0.001	0.940 (97)	0.656 (99)	0.582 (99)	-0.012 (99)	0.01	97	92	75
0.01	0.940 (97)	0.656 (99)	0.582 (99)	-0.012 (99)	0.05	97	90	73
0.05	0.940 (97)	0.656 (99)	0.582 (99)	-0.012 (99)				
0.1	0.942 (96)	0.668 (98)	0.584 (98)	-0.012 (98)				

Statistical. The rank tests are descriptive, not the pre-registered production superiority test. Density, neural, Forest-Flow, and BeyondArena stay unpooled. A small retention p -value does not flip the null scalar.

8 DISCUSSION

CTGAN and TVAE do not match the CatBoost retention, and their successful size- $4n$ artifacts miss the cap. None of these comparisons emits MFS-v2, MFS-v3, PTF-v1, or a release at the 10,240-byte tier. Table 5 is a retention table and does not emit those scores. A small Wilcoxon p -value does not flip that null. The 8,192-step budget moves the CatBoost median by 0.034 and is not the displayed model. BeyondArena fits 5 further families inside the cap and does not show retention. The unmeasured publications and the null release scores are unchanged from Section 1.2. An equal tuning budget across generators is not measured.

9 CONCLUSION

On 97 regression lineages and fit seed 11, generator `features12_steps2048` has CatBoost retention 0.940 (interval 0.914 to 0.976). CatBoost separates it from the density comparators. The linear auditor does not separate it from the Gaussian copula, and that result is not an equivalence. The 6-lineage Forest-Flow block leads on CatBoost (1/0/5) and the MLP (0/0/4). Those counts are not a certified win. The Kumo large, Mitra, and TabICL retentions are in Table 5 and are not pooled with CatBoost. MFS-v2 and MFS-v3 stay null, as do the scores in Section 1.2. Official tests stay sealed. The 10,240-byte cap remains a release gate: 98/100 displayed artifacts are inside it, and a retention does not close the gate.

AVAILABILITY

Jankurai is the repository’s conformance auditor. A passing audit means the build, tests, and boundary proofs cleared the release floor with no high finding. It is not a utility score, not an MFS value, and not a privacy certificate.

REFERENCES

Martin Abadi, Andy Chu, Ian Goodfellow, H. Brendan McMahan, Ilya Mironov, Kunal Talwar, and Li Zhang. Deep learning with differential privacy. In *Proceedings of the 2016 ACM SIGSAC Conference on Computer and Communications Security*, pp. 308–318, 2016. doi: 10.1145/2976749.2978318.

- Ahmed Alaa, Boris Van Breugel, Evgeny S. Saveliev, and Mihaela van der Schaar. How faithful is your synthetic data? sample-level metrics for evaluating and auditing generative models. In *Proceedings of the 39th International Conference on Machine Learning*, volume 162 of *Proceedings of Machine Learning Research*, pp. 290–306. PMLR, 2022. doi: 10.48550/ARXIV.2102.08921. URL <https://proceedings.mlr.press/v162/alaa22a.html>.
- Alessio Benavoli, Giorgio Corani, Janez Demšar, and Marco Zaffalon. Time for a change: a tutorial for comparing multiple classifiers through bayesian analysis. *Journal of Machine Learning Research*, 18(77):1–36, 2017. URL <https://jmlr.org/papers/v18/16-305.html>.
- Vadim Borisov, Kathrin Seßler, Tobias Leemann, Martin Pawelczyk, and Gjergji Kasneci. Language models are realistic tabular data generators. In *International Conference on Learning Representations*, 2023. doi: 10.48550/ARXIV.2210.06280. URL <https://openreview.net/forum?id=cEygmQNOeI>.
- Vadim Borisov, Tobias Leemann, Kathrin Seßler, Johannes Haug, Martin Pawelczyk, and Gjergji Kasneci. Deep neural networks and tabular data: A survey. *IEEE Transactions on Neural Networks and Learning Systems*, 35(6):7499–7519, 2024. doi: 10.1109/tnnls.2022.3229161.
- Kuntai Cai, Xiaoyu Lei, Jianxin Wei, and Xiaokui Xiao. Data synthesis via differentially private markov random fields. *Proceedings of the VLDB Endowment*, 14(11):2190–2202, 2021. doi: 10.14778/3476249.3476272.
- N. V. Chawla, K. W. Bowyer, L. O. Hall, and W. P. Kegelmeyer. SMOTE: Synthetic minority over-sampling technique. *Journal of Artificial Intelligence Research*, 16:321–357, 2002. doi: 10.1613/jair.953.
- C. K. Chow and C. N. Liu. Approximating discrete probability distributions with dependence trees. *IEEE Transactions on Information Theory*, 14(3):462–467, May 1968. doi: 10.1109/TIT.1968.1054142.
- Janez Demšar. Statistical comparisons of classifiers over multiple data sets. *Journal of Machine Learning Research*, 7(1):1–30, 2006. URL <https://jmlr.org/papers/v7/demsar06a.html>.
- Cynthia Dwork and Aaron Roth. The algorithmic foundations of differential privacy. *Foundations and Trends in Theoretical Computer Science*, 9(3–4):211–487, 2014. doi: 10.1561/04000000042.
- Nick Erickson, Lennart Purucker, Andrej Tschalzev, David Holzmüller, Prateek Mutalik Desai, David Salinas, and Frank Hutter. TabArena: A living benchmark for machine learning on tabular data. In *The Thirty-ninth Annual Conference on Neural Information Processing Systems Datasets and Benchmarks Track*, pp. 17285–17350, 2026. doi: 10.52202/085713-0519. URL <https://openreview.net/forum?id=jZqCqpCLdU>.
- Leo Grinsztajn, Edouard Oyallon, and Gael Varoquaux. Why do tree-based models still outperform deep learning on typical tabular data? In *Advances in Neural Information Processing Systems*, volume 35, pp. 507–520, 2022. doi: 10.52202/068431-0037.
- Jamie Hayes, Luca Melis, George Danezis, and Emiliano De Cristofaro. LOGAN: Membership inference attacks against generative models. *Proceedings on Privacy Enhancing Technologies*, 2019(1):133–152, 2018. doi: 10.2478/popets-2019-0008.
- Noah Hollmann, Samuel Müller, Lennart Purucker, Arjun Krishnakumar, Max Körfer, Shi Bin Hoo, Robin Tibor Schirrmeister, and Frank Hutter. Accurate predictions on small data with a tabular foundation model. *Nature*, 637(8045):319–326, 2025. doi: 10.1038/s41586-024-08328-6.
- Alexia Jolicoeur-Martineau, Kilian Fatras, and Tal Kachman. Generating and imputing tabular data via diffusion and flow-based gradient-boosted trees. In *Proceedings of The 27th International Conference on Artificial Intelligence and Statistics*, volume 238 of *Proceedings of Machine Learning Research*, pp. 1288–1296. PMLR, 2024. doi: 10.48550/ARXIV.2309.09968. URL <https://proceedings.mlr.press/v238/jolicoeur-martineau24a.html>.

- Jayoung Kim, Chaejeong Lee, and Noseong Park. STaSy: Score-based tabular data synthesis. In *International Conference on Learning Representations*, 2023. doi: 10.48550/ARXIV.2210.04018. URL https://openreview.net/forum?id=lmNssCWt_v.
- Akim Kotelnikov, Dmitry Baranchuk, Ivan Rubachev, and Artem Babenko. TabDDPM: Modelling tabular data with diffusion models. In *Proceedings of the 40th International Conference on Machine Learning*, volume 202 of *Proceedings of Machine Learning Research*, pp. 17564–17579, 2023. doi: 10.48550/ARXIV.2209.15421. URL <https://proceedings.mlr.press/v202/kotelnikov23a.html>.
- Chaejeong Lee, Jayoung Kim, and Noseong Park. CoDi: Co-evolving contrastive diffusion models for mixed-type tabular synthesis. In *Proceedings of the 40th International Conference on Machine Learning*, volume 202 of *Proceedings of Machine Learning Research*, pp. 18940–18956. PMLR, 2023. doi: 10.48550/ARXIV.2304.12654. URL <https://proceedings.mlr.press/v202/lee23i.html>.
- Tennison Liu, Zhaozhi Qian, Jeroen Berrevoets, and Mihaela van der Schaar. GOGGLE: Generative modelling for tabular data by learning relational structure. In *International Conference on Learning Representations*, 2023. URL <https://openreview.net/forum?id=fPVRcJqspu>.
- Junwei Ma, Apoorv Dankar, George Stein, Guangwei Yu, and Anthony Caterini. TabPFGen – tabular data generation with TabPFN. In *NeurIPS 2023 Second Table Representation Learning Workshop*, 2023. URL <https://openreview.net/forum?id=4MkkNsAEmO>.
- Duncan McElfresh, Sujay Khandagale, Jonathan Valverde, Vishak Prasad C, Ganesh Ramakrishnan, Micah Goldblum, and Colin White. When do neural nets outperform boosted trees on tabular data? In *Advances in Neural Information Processing Systems*, volume 36, pp. 76336–76369, 2023. doi: 10.52202/075280-3337. URL https://proceedings.neurips.cc/paper_files/paper/2023/hash/f06d5ebd4ff40b40dd97e30cee632123-Abstract-Datasets_and_Benchmarks.html. Datasets and Benchmarks Track.
- Ryan McKenna, Gerome Miklau, and Daniel Sheldon. Winning the NIST contest: A scalable and general approach to differentially private synthetic data. *Journal of Privacy and Confidentiality*, 11(3), 2021. doi: 10.29012/jpc.778.
- Ryan McKenna, Brett Mullins, Daniel Sheldon, and Gerome Miklau. AIM: An adaptive and iterative mechanism for differentially private synthetic data. *Proceedings of the VLDB Endowment*, 15(11):2599–2612, 2022. doi: 10.14778/3551793.3551817.
- Markus Mueller, Kathrin Gruber, and Dennis Fok. Continuous diffusion for mixed-type tabular data. In *International Conference on Learning Representations*, 2025. doi: 10.48550/ARXIV.2312.10431. URL <https://openreview.net/forum?id=QPToBPn4lZ>.
- Beata Nowok, Gillian M. Raab, and Chris Dibben. synthpop: Bespoke creation of synthetic data in R. *Journal of Statistical Software*, 74(11):1–26, 2016. doi: 10.18637/jss.v074.i11.
- Randal S. Olson, William La Cava, Patryk Orzechowski, Ryan J. Urbanowicz, and Jason H. Moore. PMLB: a large benchmark suite for machine learning evaluation and comparison. *BioData Mining*, 10(1):36, 2017. doi: 10.1186/s13040-017-0154-4.
- Noseong Park, Mahmoud Mohammadi, Kshitij Gorde, Sushil Jajodia, Hongkyu Park, and Youngmin Kim. Data synthesis based on generative adversarial networks. *Proceedings of the VLDB Endowment*, 11(10):1071–1083, 2018. doi: 10.14778/3231751.3231757.
- Neha Patki, Roy Wedge, and Kalyan Veeramachaneni. The synthetic data vault. In *2016 IEEE International Conference on Data Science and Advanced Analytics (DSAA)*, pp. 399–410, 2016. doi: 10.1109/DSAA.2016.49.

- Liudmila Prokhorenkova, Gleb Gusev, Aleksandr Vorobev, Anna Veronika Dorogush, and Andrey Gulin. CatBoost: unbiased boosting with categorical features. In *Advances in Neural Information Processing Systems*, volume 31, 2018. doi: 10.48550/ARXIV.1706.09516. URL <https://papers.nips.cc/paper/2018/hash/14491b756b3a51daac41c24863285549-Abstract.html>.
- Zhaozhi Qian, Rob Davis, and Mihaela van der Schaar. Synthcity: A benchmark framework for diverse use cases of tabular synthetic data. In *Advances in Neural Information Processing Systems*, volume 36, pp. 3173–3188, 2023. doi: 10.52202/075280-0140. URL https://proceedings.neurips.cc/paper_files/paper/2023/hash/09723c9f291f6056fd1885081859c186-Abstract-Datasets_and_Benchmarks.html. Datasets and Benchmarks Track.
- Gillian M. Raab, Beata Nowok, and Chris Dibben. Practical data synthesis for large samples. *Journal of Privacy and Confidentiality*, 7(3):67–97, 2018. doi: 10.29012/jpc.v7i3.407.
- Joseph D. Romano, Trang T. Le, William La Cava, John T. Gregg, Daniel J. Goldberg, Praneel Chakraborty, Natasha L. Ray, Daniel Himmelstein, Weixuan Fu, and Jason H. Moore. PMLB v1.0: an open-source dataset collection for benchmarking machine learning methods. *Bioinformatics*, 38(3):878–880, 2022. doi: 10.1093/bioinformatics/btab727.
- Juntong Shi, Minkai Xu, Harper Hua, Hengrui Zhang, Stefano Ermon, and Jure Leskovec. TabDiff: a mixed-type diffusion model for tabular data generation. In *International Conference on Learning Representations*, 2025. doi: 10.48550/ARXIV.2410.20626. URL <https://openreview.net/forum?id=swvURjrt8z>.
- Reza Shokri, Marco Stronati, Congzheng Song, and Vitaly Shmatikov. Membership inference attacks against machine learning models. In *2017 IEEE Symposium on Security and Privacy (SP)*, pp. 3–18, 2017. doi: 10.1109/sp.2017.41.
- Ravid Shwartz-Ziv and Amitai Armon. Tabular data: Deep learning is not all you need. *Information Fusion*, 81:84–90, 2022. doi: 10.1016/j.inffus.2021.11.011.
- Joshua Snoke, Gillian M. Raab, Beata Nowok, Chris Dibben, and Aleksandra Slavkovic. General and specific utility measures for synthetic data. *Journal of the Royal Statistical Society Series A: Statistics in Society*, 181(3):663–688, 2018. doi: 10.1111/rssa.12358.
- Aivin V. Solatorio and Olivier Dupriez. REaLTabFormer: Generating realistic relational and tabular data using transformers, 2023.
- Boris van Breugel, Hao Sun, Zhaozhi Qian, and Mihaela van der Schaar. Membership inference attacks against synthetic data through overfitting detection. In *Proceedings of The 26th International Conference on Artificial Intelligence and Statistics*, volume 206 of *Proceedings of Machine Learning Research*, pp. 3493–3514. PMLR, 2023. doi: 10.48550/ARXIV.2302.12580. URL <https://proceedings.mlr.press/v206/breugel23a.html>.
- David S. Watson, Kristin Blesch, Jan Kapar, and Marvin N. Wright. Adversarial random forests for density estimation and generative modeling. In *Proceedings of the 26th International Conference on Artificial Intelligence and Statistics*, volume 206 of *Proceedings of Machine Learning Research*, pp. 5357–5375, 2023. doi: 10.48550/ARXIV.2205.09435. URL <https://proceedings.mlr.press/v206/watson23a.html>.
- Lei Xu, Maria Skoularidou, Alfredo Cuesta-Infante, and Kalyan Veeramachaneni. Modeling tabular data using conditional GAN. In *Advances in Neural Information Processing Systems*, volume 32, 2019. doi: 10.48550/ARXIV.1907.00503. URL <https://papers.nips.cc/paper/2019/hash/254ed7d2de3b23ab10936522dd547b78-Abstract.html>.
- David Turtora Zagardo. Geometry-aware tabular diffusion. In *Proceedings of the 43rd International Conference on Machine Learning*, volume 306 of *Proceedings of Machine Learning Research*, pp. 152812–152835. PMLR, 2026. doi: 10.48550/ARXIV.2606.02607. URL <https://proceedings.mlr.press/v306/zagardo26a.html>.

- Hengrui Zhang, Jiani Zhang, Balasubramaniam Srinivasan, Zhengyuan Shen, Xiao Qin, Christos Faloutsos, Huzefa Rangwala, and George Karypis. Mixed-type tabular data synthesis with score-based diffusion in latent space. In *International Conference on Learning Representations*, 2024. doi: 10.48550/ARXIV.2310.09656. URL <https://openreview.net/forum?id=4Ay23yeuz0>.
- Jun Zhang, Graham Cormode, Cecilia M. Procopiuc, Divesh Srivastava, and Xiaokui Xiao. PrivBayes: Private data release via bayesian networks. In *Proceedings of the 2014 ACM SIGMOD International Conference on Management of Data*, pp. 1423–1434, 2014. doi: 10.1145/2588555.2588573.
- Kevin Alex Zhang, Kalyan Veeramachaneni, and Carles Sala. SDMetrics: Synthetic data metrics. Open-source library, MIT Data To AI Lab, 2020. URL <https://docs.sdv.dev/sdmetrics/>.
- Zilong Zhao, Aditya Kunar, Robert Birke, and Lydia Y. Chen. CTAB-GAN: Effective table data synthesizing. In *Proceedings of The 13th Asian Conference on Machine Learning*, volume 157 of *Proceedings of Machine Learning Research*, pp. 97–112. PMLR, 2021. URL <https://proceedings.mlr.press/v157/zhao21a.html>.
- Zilong Zhao, Aditya Kunar, Robert Birke, Hiek Van der Scheer, and Lydia Y. Chen. CTAB-GAN+: Enhancing tabular data synthesis. *Frontiers in Big Data*, 6, 2024. doi: 10.3389/fdata.2023.1296508.